Name: Shaunak Karnik

Tigran Hakobyan

② Let's express the problem in linear programming terms.

For each edge $\langle i,j \rangle$ of the graph we define variable x_{ij} such that if $x_{ij}=1$ then the $\langle i,j \rangle$ edge is going to be a part of the bipartite matching, if $x_{ij}=0$, it means that the edge is not in the matching set.

Our purpose is to include as many edges as possible. As a function we will have

(1) $\boxed{F=\sum x_{ij} \rightarrow \text{maximize}}$, where x_{ij} is an edge connecting i,j vertices

We should try to maximize this function satifying the condition that there is no connected edges in the matching set.

(2) $\boxed{\sum_{e\in N(v)} x_e \leq 1}$ for $\forall v\in V$ & $x_e \geq 0$ $\forall e\in E$, where $N(v)$ is the set of edges which have v as a vertex.

To combine them,

$$\boxed{\begin{array}{l} \sum x_{ij} \rightarrow \text{maximize} \\ \sum_{e\in N(v)} x_e \leq 1, \ \forall v\in V \ \& \ x_e \geq 0 \ \ \forall e\in E \end{array}}$$

SHAUNAK KARNIK TIGRAN HAKOBYAN

Problem 3:

* The decision problem is in NP since a certificate for instance of problem is subset of edges that forms an s-t path of total weight. Checking whether a set of edges exceed k is obviously possible in polynomial time.

Shortest path with -ve weights is NP-complete.

Let $G = (V, E)$ be instance of Hamiltonian path problem.
Construct from G an instance $G' = (V', E')$. [V: vertices, E: edges]
Transform network with -1 wt. on each edge.
Let k be set of $|V| - 1$.
Constructing k is counting no. of vertices in V taking $O(n)$ time.
~~Specifying s & t~~. Constructing V' from V takes $O(1)$ time.
Constructing E' requires replacing each edge in E by 2 directed edges $\rightarrow$ adding two additional edges at start and end. This takes $O(m+n)$ time.

G is a YES instance of Hamiltonian Path if & only if G' is a YES instance of shortest path s-t of length $(-k)$.

Let $((v_1, v_2) \ldots (v_{n-1}, v_n))$ be Hamiltonian Path for G. ~~So, it is a simple s-t path of weight |V|-1 for G'~~.
G' is formed by adding addition edges like $((s\ v_1)(v_1, v_2) \ldots (v_{n-1}, v_n)(v_n, t))$ of weight $|V| - 1$ for G'. So G' is a YES instance of ~~shortest~~ path problem.
So it is a simple s-t path of weight

Conversely, suppose a path $((s, v_1)(v_1, v_2) \ldots (v_{x-1}\ v_x)(v_x\ t))$ for some x) is a simple path of weight $|V| - 1$. The edges (s, v_1) & (v_x, t) have zero weight so weight for graph G is also $|V| - 1$. Since all edges are not connected to s or t, $x = n$. Since $(v_1\ v_2)(v_2, v_3) \ldots (v_{x-1}\ v_x)$ is a simple directed path of $n-1$ edges in G', it is also a simple undirected path of $n-1$ edges in G. So if the shortest path has $|V| - 1$ length then this is a Hamiltonian path.

Reference: Utrecht University [Mastermath & LNMB course]
wwwhome.math.utwente.nl/~keirnw/lectures/DO/Re-Exam-Sol.pdf